that its surface is young, because bodies whose surfaces don't keep changing have impact craters (like Earth's Moon). A surface change can happen due to plate tectonics and volcanism where the inner material of the planet or moon comes out and forms the new, smooth outer layer.

Europa does indeed have its material from inside forming newer layers of smooth surface outside due to plate tectonics. On Earth, tectonics occur because the core is hot and molten. On Europa, tectonics occur because of tidal flexing. Jupiter and the other large moons, (called Ganymede and Callisto) constantly tug at Europa, causing it to be agitated internally. This releases heat and provides geothermal energy. This discovery of energy also requires SIPS to reexamine the definition and usage Goldilocks zone.

There is evidence of a subsurface liquid ocean of water under the icy crust. The heat from the geothermal activity keeps the water in liquid state. Europa has a magnetic field that is induced by Jupiter's. Furthermore, hydrogen peroxide is abundant on the surface. As hydrogen peroxide decays to molecular oxygen, it could provide necessary support to life. If there is any life, it is definitely restricted to aquatic forms.

- Energy: Adequate
- Suitable Gases: Oxygen
- Magnetic Field: Yes
- Liquid Water: Abundant
- HML: 4.3

GS06.13 (Enceladus)

Enceladus : This is an icy moon of the ringed planet, coded GS06 (translation: Saturn), and is barely bigger than a regular asteroid. It is a suitable candidate for our study for the same reasons as Europa. It is a rocky-ice body with a surface of nearly smooth, fresh, bright water ice. The satellite's biggest appeal is cryovolcanism. Unlike volcanoes on Earth, the ones on Enceladus spew out ice, water, salt, and dust. They are called Plumes.

Just like Europa, Enceladus's heat comes from tidal flexing, but of a different kind. Enceladus is tidally locked to Saturn in a 1:1 resonance, which means that the same side of the satellite faces the planet and one revolution equals one rotation. Therefore, one side of Enceladus constantly faces more gravitational tug from Saturn than the other side, producing a lot of tidal forces that give rise to subsurface and atmospheric disturbances, causing heat and geological activity.

Naturally, this heat keeps a global subsurface ocean of water in the liquid form and also causes cryovolcanism. Enceladus even contains organic carbon compounds and lies within Saturn's magnetic field. Any life would be confined to the subsurface ocean.

Not very hospitable but Morkonians have been known to survive in worse

- Energy: Adequate
- Suitable Gases: Methane and water vapour
- Magnetic Field: Yes
- Liquid Water: Abundant
- HML: 3.9

GS06.06 (Titan)

Titan is another satellite of Saturn and is the second largest satellite in the solar system. It is bigger than even the first planet in the system. It is the only satellite to contain a thick atmosphere. This atmosphere contains nitrogen for the most part, but it also contains clouds of methane that rain down on the moon. Titan has a full methane cycle like the GS03 water cycle, complete with weather, erosion, wind, and associated surface features. It is the only other body that has can sustain liquid on its surface.

It sits within Saturn's magnetic field for the most part of its orbit and obtains energy due to tidal flexing from Saturn. It is a prebiotic body, meaning it has the perfect conditions for life to form. Our probes have showed conditions similar to theorised early Earth, the primordial soup from which life formed there.

Titan receives a low ranking on the FHM because exploration and study is not in line with looking for carbon-based life forms. Life on Titan would be unusual and exotic, if existent. But primarily, it would be unidentifiable. Organisms that could thrive in the methane lakes would inhale oxygen and exhale methane. However, proposals have been submitted for future studies of Titan as the ongoing expansion of Sun will eventually make Titan habitable enough to sustain liquid water within a few million years.

- Energy: Adequate
- Suitable Gases: Methane
- Magnetic Field: Yes
- Liquid Water: Possibly sub surface, no evidence
- HML: 2.5

It is our strong recommendation to the Astrosciences Committee at SIPS that the study of this star system continue due to highly favorable conditions on G3. Probes to be dispatched at the earliest.

The above transmission was received from the Chi Sagittarii group of stars in the region of the Sagittarius constellation. Since the successful interpretation of the signal this region of deep space has been under regular scanning but so far, no other signal has been detected. Even more surprising is the fact that none of our telescopes have been able to detect exoplanets in this region, prompting scientists to think that this was probably a message transmitted accidentally in the wrong direction by a moving ship.